Align-for-Fusion: Harmonizing Triple Preferences via Dual-oriented Diffusion for Cross-domain Sequential Recommendation

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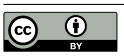
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Abstract

Personalized sequential recommendation aims to predict the appropriate items to users from their behavioral sequences. To alleviate the data sparsity and interest drift issues, conventional approaches typically utilize the additional behaviors from other domains via cross-domain transition. However, existing cross-domain sequential recommendation (CDSR) algorithms follow the *align-then-fusion* paradigm which conducts the representation-level alignment across multiple domains and mechanically combine them for recommendation, overlooking the fine-grained multi-domain fusion. Inspired by the advancements of diffusion models (DMs) in distribution matching, we propose an *align-for-fusion* framework for CDSR to **H**arm**o**nize t**ri**ple preferences utili**z**ing Dual-**o**rie**n**ted DMs (HorizonRec). Specifically, we first investigate the uncertainty injection of DMs and attribute the fundamental factor of the instability in existing DMs recommenders to the stochastic noise and propose a Mixed-conditioned Distribution Retrieval strategy which leverages the retrieved distribution from users' authentic behavioral logic as a bridge across the triple domains, enabling consistent multi-domain preference modeling. To suppress the potential noise and emphasize target-relevant interests during multi-domain user representation fusion, we further propose a Dual-oriented Preference Diffusion method to guide the extraction of preferences aligned with users' authentic interests from each domain under the supervision of the mixed representation. We conduct extensive experiments and analyses on four CDSR datasets from two distinct platforms to verify the effectiveness and robustness of our HorizonRec and its effective mechanism in fine-grained fusion of triple domains. Our code and datasets are available in https://github.com/YongfuZha/HorizonRec.

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CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Recommender System, Cross-domain Sequential Recommendation, Diffusion Models

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1 Introduction

Recommender systems (RS) play a critical role in mitigating information overload [28, 40], with applications such as social recommendation [38, 39] and fairness-aware recommendation [29, 34]. As user behaviors increasingly exhibit temporal patterns, sequential recommendation (SR) has become crucial for modeling evolving user interests and delivering context-aware recommendations [12, 21, 36]. However, the data sparsity problem and interest drift issue inherent in SR hinder its temporal preference understanding, ultimately limiting its performance [41]. Recently, cross-domain sequential recommendation (CDSR) has attracted increasing attention, as it can leverage users' sequential behaviors from auxiliary domains to enrich target domain and infer the dynamic preferences via interest shifts observed elsewhere [2, 10, 19, 20].

Prior studies typically focus on the knowledge transfer from source to target domain, particularly in cross-domain share-account recommendation [5, 24]. Recently, a series of studies highlight the significance of mixed domain which encompasses user behaviors from both source and target domain and facilitate preference modeling via temporal alignment [1] and multi-granular correlation modeling [2, 20]. As illustrated in Figure 1 (a), these works primarily follow the *align-then-fusion* paradigm to separate the representation-level alignment across multiple domains with the subsequent mechanical combination. To verify this issue, we also conduct the

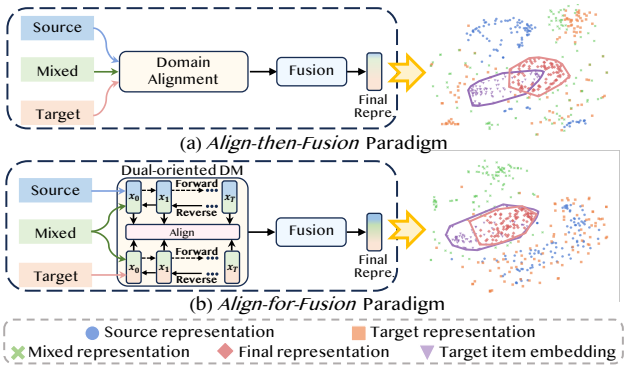


Figure 1: Comparison of the illustration and the t-SNE visualization on user representations and target item embeddings on Music→Book setting between the *align-then-fusion* and *align-for-fusion* paradigms. Our method yields better alignment and target-aware representation.

t-SNE visualization on Tri-CDR [20] with 100 randomly-selected and their domain-specific representations, final representations, and target item embeddings in the inference phase. We can observe the noticeable overlap and intersection among representations from triple domains, yet the distribution of final representation exhibits a significant gap from that of target items—an inevitable consequence of the *align-then-fusion* paradigm. It may inadvertently introduce noise interference from other domains or discard critical domain-specific signals, resulting in the suboptimal alignment with users' true interest distributions in target domain. Inspired by the remarkable performance of Diffusion Models (DMs) in Computer Vision and Natural Language Processing, some researchers have initiated a line of pioneering works to explore their potential in recommendation. For instance, DiffuRec [15] and DreamRec [41] successfully inject the uncertainty modeling capabilities of DMs into sequential recommendation, but their focus remains limited to intra-domain settings, making them less applicable to cross-domain scenarios. More recently, several cross-domain recommendation endeavors (*i.e.,* DiffRec [32] and DMCDR [14]) utilize DMs to capture the inter-domain collaborative correlations, overlooking the sequential patterns inherent in user behaviors. Therefore, when directly applied to CDSR tasks that require joint modeling of cross-domain and temporal dependencies, they yield suboptimal performance.

To alleviate these issues, we propose an *align-for-fusion* framework, HorizonRec, which regards mixed domain as a unified Horizon to break through the boundaries of cross-domain preference modeling, enabling Dual-oriented DM to precisely harmonize user preferences across triple behavioral landscapes. Specifically, we begin by analyzing the noise interference and information absence issues in *align-then-fusion* paradigm, analyze the critical factor of instability in existing DM-based recommenders and attribute it to the deviation introduced by random noise. Motivated by the core of Retrieval-Augmented Generation, we propose a Mixed-Conditioned Distribution Retrieval (MDR) module to retrieve distributions most relevant to users' global interests based on their authentic behavioral logic. These retrieved signals serve as the initial distribution for the subsequent diffusion process to ensure that user preference representations remain globally aligned throughout diffusion, thereby facilitating the proposed *align-for-fusion* paradigm. Then we propose a Dual-oriented Preference Diffusion (DPD) module to simultaneously conduct diffusion on source and target domains under the guidance of the mixed domain, enabling the extraction of domain-specific knowledge that is closely aligned with the user's global preference and target-domain interest. As illustrated in Figure 1 (b), the final representation is entirely located within the distributional region of target item embedding in HorizonRec, demonstrating the effectiveness of our *align-for-fusion* paradigm. It enables dual-oriented diffusion to serve as a horizon for harmonizing the aligned preferences across triple domains, ultimately yielding a unified representation of the users' personalized interests.

We have conducted extensive experiments on four CDSR settings from two diverse platforms to demonstrate the superiority of our HorizonRec. We also conduct various ablation studies, parameter sensitive analysis, computational complexity analysis and visualization to verify the effectiveness of our proposed MDR and DPD. The contributions are summarized as follows:

- We have delicately verified the significance of *align-for-fusion* paradigm for target-related knowledge extraction and fine-grained triple preferences harmonizing. To the best of our knowledge, we are among the pioneer to investigate this paradigm via dual-oriented diffusion models in CDSR.
- We design a Mixed-conditioned Distribution Retrieval module which retrieves the globally-focused distribution to support diffusion process for stable multi-domain preference modeling.
- We conduct extensive experiments and analyses to demonstrate the effectiveness of each component within our HorizonRec on various datasets.

2 Related Work

2.1 Sequential Recommendation

Sequential Recommendation (SR) aims to model the users' dynamic preference pattern with the historical behavior sequences and recommend the appropriate item to him/her [17]. Early SR approaches attempt to infer users' temporal preference with the Markov Chains [6, 26], Convolution Neural Network [31], and Recurrent Neural Network [7]. With the success of the self-attention mechanism, SASRec [12] leverages Transformer as the sequential encoder to model users' short- and long-term preferences. CL4SRec [36] designs several contrastive learning tasks to achieve mutual information maximization on sequence level. Recently, PDRec [18], DiffuRec [15], and DreamRec [41] utilize the advanced diffusion models to inject the uncertainty into user preference modeling in a generative manner to improve its robustness and generalization. However, the research emphasis of these studies remains confined to uncovering the intra-domain dynamic preference patterns, hindering their effectiveness when directly applied to cross-domain scenarios.

2.2 Cross-domain Recommendation

Prior cross-domain recommendation (CDR) studies [13, 33, 42, 43] aim to enhance target-domain performance by leveraging auxiliary-domain information. For instance, SSCDR [11] introduces a semi-supervised mapping method to transfer preferences across domains. UniCDR [3] employs domain masking and contrastive learning to extract domain-invariant representations. Recently, some works

start introducing diffusion models into CDR, where DiffCDR[37] and DMCDR [14] leverage it to model preference uncertainty and enhance cross-domain alignment. Extending to cross-domain sequential recommendation (CDSR), Tri-CDR [20] designs a triple contrastive objective to capture fine-grained correlations among triple domains. SyNCRec [25] recognize the harmful gradients to mitigate negative transfer, and ABXI [1] extracts domain-invariant components for adaptive integration into domain-specific representations. However, these works mechanically align inter-domain user behaviors via tailored optimization objectives, while overlooking the nuanced correlations and distinctions among multi-domain features during the fusion phase.

2.3 Diffusion Models for Recommendation

Motivated by the remarkable achievement of Diffusion Models (DMs) in image synthesis [4, 27] and categorical data modeling [9], some researchers attempt to incorporate it into recommendation [16, 22, 35]. For instance, DiffRec [32] models the generative process of user interactions via denoising. MCDRec [23] introduces a multi-channel diffusion process to enhance user-item semantic alignment. To further investigate its distribution matching property, recent studies attempt to apply it in SR and CDR tasks, which focus more on evolving user preferences. DiffuRec [15] generates future behaviors through a diffusion decoder, guided by a preference-oriented reverse process. DreamRec [41] perturbs temporal sequences with noise and leverages conditional diffusion to generate denoised behavior sequences. For the more specific CDR tasks, the pioneering DiffCDR [37] utilize DM for multi-domain preference sharing, incorporating auxiliary-domain behaviors during decoding. DMCDR [14] further proposes a preference-guided diffusion strategy that conditions the generation on cross-domain preferences, improving representations for cold-start users. However, existing works focus either on the multi-modality of items or the temporal preference shifts within a single domain, while rarely model the cross-domain and sequential dependencies from users' cross-domain behaviors.

3 Preliminary

Inspired by the powerful distribution matching ability of Denoising Diffusion Probabilistic Models (DDPM) [8], we adopt it as the diffusion backbone for our HorizonRec.

Forward Process. DDPM defines a pre-specified Markov process that progressively adds Gaussian noise to a clean input $\mathbf{x}_0$, such that the final state $\mathbf{x}_T$ approximates an isotropic Gaussian distribution: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\,\mathbf{x}_{t-1}, \beta_t\mathbf{I}). \tag{1}$$

By unfolding the forward process, the diffusion state distribution at time step t can be formulated as:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \text{ where } \quad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s). \tag{2}$$

Through reparameterization technology, $\mathbf{x}_t$ can be represented as $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Reverse Process. Regarding the reverse process, we utilize the original representation $\mathbf{x}_0$ as the reverse target to generate the estimated representation $\hat{\mathbf{x}}_0$ via a estimator $f_\theta(.)$ which is conditioned on the noised input $\mathbf{x}_t$ and the additional side information:

$$\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_t, \mathbf{c}, t) = f_\theta(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \mathbf{c}, t), \tag{3}$$

where $\mathbf{c}$ denotes the specific contextual guidance (e.g., the mixed-domain representation in our HorizonRec).

Training Objective. To align with the reverse process, we use the Mean-Squared Error (MSE) loss function on the original representation $\mathbf{x}_0$ and the estimated representation $\hat{\mathbf{x}}_0$ to optimize the estimator $f_\theta(.)$:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}, t}\left[\left\|\mathbf{x}_0 - f_\theta\left(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \mathbf{c}, t\right)\right\|^2\right]. \tag{4}$$

This formulation enables stable training and encourages the generative model to recover semantically meaningful user representations guided by cross-domain context.

4 Methodology

4.1 Problem Formulation

CDSR aims to predict users' next interaction in the target domain by exploiting dependencies within and across domains. To achieve this, we first define the *source behavioral sequence* as $S_u^S = (i_1^S, i_2^S, \ldots, i_p^S)$ and *target behavioral sequence* as $S_u^T = (i_1^T, i_2^T, \ldots, i_q^T)$ of user u, where p and q denote the sequence lengths of source domain S and target domain T, $i_k^S \in \mathcal{I}^S$ and $i_j^T \in \mathcal{I}^T$ represent the k-th and the j-th behavior in S_u^S and S_u^T, respectively. Then we construct the *mixed behavioral sequence* $S_u^M = \left(i_1^M, i_2^M, \ldots, i_{p+q}^M\right)$ by chronologically merging S_u^S and S_u^T. Here, CDSR attempts to maximize the probability of the next behavior i_{q+1}^T of user u in target domain according to his/her historical sequences S_u^S, S_u^M, and S_u^T, which can be formulated as:

$$\arg\max_{i_{q+1}^T \in \mathcal{I}^T} p(i_{q+1}^T \mid S_u^S, S_u^T, S_u^M). \tag{5}$$

4.2 Overall Framework

In this section, we introduce a *align-for-fusion* framework, HorizonRec, which aims to harmonize triple domains via dual-oriented diffusion to explicitly capture and integrate users' preferences from the source, target, and mixed domains. As illustrated in Figure 2, our HorizonRec includes two key modules: i) Mixed-conditioned Distribution Retrieval, which aims to dynamically regulate the noise distribution in the diffusion process by incorporating historical behaviors from mixed domain that are close to users' global preferences. By introducing structured external priors, it controls the direction and magnitude of noise injection through similar sequences, suppressing irrelevant domain information and enhancing the accuracy of CDSR. ii) Dual-oriented Preference Diffusion: We introduce a dual-oriented diffusion models on source and traget domain simultaneously, harmonizing users' local interests with the global preference strategy and facilitating joint modeling of user preferences across triple domains.

4.3 Base Recommender

CDSR aims to predict users' next behavior in the target domain according to their historical behavior sequences from other domains, relying on the accurate modeling of temporal behavior patterns

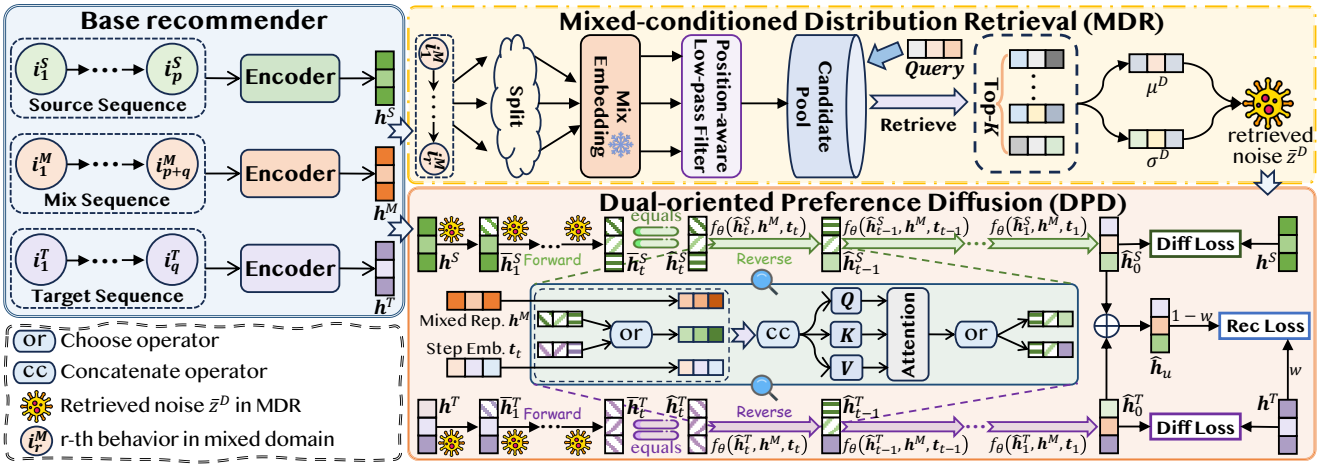


Figure 2: The overview of our proposed HorizonRec framework.

within each domain. To achieve this, we employ SASRec [12] as our domain-specific recommender. Taking the target domain T as an example, we first map the behavioral sequence $S_u^T = (i_1^T, i_2^T, \ldots, i_q^T)$ into the embedding sequence $\mathbf{E}_u^T = (\mathbf{e}_1^T, \mathbf{e}_2^T, \ldots, \mathbf{e}_q^T)$ using the embedding matrix $\mathbf{E}^T \in \mathbb{R}^{|\mathcal{I}^T| \times d}$. Then we utilize the self-attention mechanism and pointwise feed-forward network to obtain the hidden sequence $\boldsymbol{H}_u^T$:

$$\text{Attn}_u^T = \text{Softmax}\left(\boldsymbol{Q}^T (\boldsymbol{K}^T)^\top / \sqrt{d}\right) \boldsymbol{V}^T. \tag{6}$$

where the hidden sequence is computed as $\boldsymbol{H}_u^T = \text{ReLU}(\text{Attn}_u^T \boldsymbol{W}_1 + \boldsymbol{b}_1)\boldsymbol{W}_2 + \boldsymbol{b}_2$, $\boldsymbol{Q}^T, \boldsymbol{K}^T, \boldsymbol{V}^T$ are projected from the embedding sequence $\mathbf{E}_u^T$, $\boldsymbol{W}_1, \boldsymbol{W}_2, \boldsymbol{b}_1, \boldsymbol{b}_2$ are weights and biases, respectively. The hidden sequences $\boldsymbol{H}_u^S$ and $\boldsymbol{H}_u^M$ of user u on the source and mixed domains are similarly constructed through this process. Following the representative CDSR method [20], we utilize SASRec (S+T) as our base cross-domain sequential recommender. Specifically, we concatenate the sequence-level representations $\boldsymbol{h}_u^S$ and $\boldsymbol{h}_u^T$ of source and target domains to generate the final user representation $\boldsymbol{h}_u$ via a multi-layer perceptron (MLP) [1].

4.4 Mixed-conditioned Distribution Retrieval

Conventional DM-based recommenders typically inject user-agnostic Gaussian noise into user behaviors [32] or item representations [23], where the controlled noise can facilitate the smoother representations modeling, thereby partially alleviating data sparsity. However, such purely random noise may disrupt subtle cross-domain correlations, exacerbate distributional shifts between domains, and obscure the consistency of user preferences across triple domains in CDSR tasks. To address this issue, we propose the Mixed-conditioned Distribution Retrieval (MDR) module, which constructs behavior-aligned and user-specific noise from historical mixed-domain interactions to support the following diffusion process within our HorizonRec.

Given the mixed-domain sequence $S_u^M = (i_1^M, i_2^M, \ldots, i_{|S_u^M|}^M)$ of user u, we firstly extract all the contiguous subsequences ending with target-domain items to form the retrieval candidate:

$$\mathcal{D}_u = \left\{ (i_1^M, \ldots, i_l^M) \mid 1 < l \le |S_u^M|,\ i_l^M \in \mathcal{I}^T \right\}. \tag{7}$$

[1]To reduce notational clutter, we omit user u in the following sections for simplicity.

Then, we extract the representations of each item within the subsequence $d \in \mathcal{D}_u$, which is a sequence in the mixed domain ending with a target-domain item and formed by combining user preferences from both the source and target domains. Therefore, the start and each end point of d correspond to the user's first interaction and each target-domain interaction, respectively. These item representations are retrieved from the pre-trained item embedding matrix. We then apply a position-aware low-pass filter [30] to highlight the more recent behaviors in candidate embeddings $\mathbf{e}_d$.

$$\mathbf{e}_d = \sum_{j=k}^{l} \left(c - 1/(1 + \left(\frac{p(i_j)}{l - j + 1} \right)^n) \right) \mathbf{e}(i_j^M), \tag{8}$$

where $p(i_j)$ denotes the position of item i_j, c controls the initial value of weight of $\mathbf{e}(i_j^M)$, and n represents the decay rate. According to Equation 8, we can extract all the filtered embeddings from $\mathcal{D}_u$ of user u to construct the user-specific embedding matrix E_u. Applying this procedure across all users results in the construction of a global retrieval database as $\mathcal{D} = \bigcup_{u \in \mathcal{U}} E_u^d$.

For each sequence, we utilize the user's final representation $\boldsymbol{h}^D$ ($D \in S, T$) from the source and target domains to retrieve the top-K most similar segments from $\mathcal{D}$ using cosine similarity, denoted as $\mathbf{d}_1^D, \mathbf{d}_2^D, \ldots, \mathbf{d}_K^D$. These segments serve as semantic anchors for constructing the enhanced noise distribution as follows:

$$\bar{z}^D = \mu^D + \sigma^D \xi, \quad \xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{9}$$

where $\mu^D = \frac{1}{K} \sum_{i=1}^{K} (\mathbf{d}_i - \boldsymbol{h}^D)$ and σ^D denote the mean and standard deviation across all the retrieved segments. And the retrieved noise $\bar{z}^D$ is sampled based on the retrieved segments from this user historical behaviors under the specific domain D.

Theoretical Properties. MDR provides behaviorally aligned noise with two desirable properties:

Lemma 4.1 (Retrieval Distribution Structure). *The retrieval distribution from MDR satisfies:*

$$\bar{z}^D = \mu^D + \sigma^D \xi, \quad \textit{with} \quad {\sigma^D}^2 \le \sigma_{std}^2, \tag{10}$$

In expectation, and μ^D points toward target-ending trajectories.

Proposition 4.2 (Dual-Oriented Property). *Under the smoothness of the specific CDSR task:*

$$\mathbb{E}[p(i_{q+1}^T|\boldsymbol{h}^D + \bar{z}^D, \boldsymbol{h}^M)] \geq \mathbb{E}[p(i_{q+1}^T|\boldsymbol{h}^D + \bar{z}_{std}, \boldsymbol{h}^M)]. \quad (11)$$

where $\bar{z}_{std} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ denotes the standard Gaussian noise.

These analyses support the intuition behind the design of MDR, showing that the retrieved noise $\bar{z}^D$ not only provides dual-oriented guidance, but also exhibits lower variance than standard Gaussian perturbations in CDSR tasks [2].

4.5 Dual-oriented Preference Diffusion

CDSR assumes that user behaviors from multiple domains reflect diverse and dynamic interests, providing valuable auxiliary signals for target-domain modeling. However, indiscriminately leveraging all behaviors may introduce redundancy or noise, undermining recommendation performance. To mitigate this, we propose a Dual-oriented Preference Diffusion (DPD) module on the basic of the enhanced noise distribution in MDR, which leverages the generative power of diffusion models to uncover shared user intents. It injects retrieval-enhanced noise into source and target domain representations and utilizes mixed-domain preferences as a semantic bridge to guide cross-domain interest modeling and harmonize user preferences among triple domains.

Dual-oriented diffusion. Zooming into the implementation details of existing DM-based cross-domain recommenders, we observe that these approaches typically condition the diffusion process of target-domain representations on source-domain information, while neglecting explicit modeling of user behavioral representations in the source domain. Notably, source-domain behavioral sequences also encode informative dynamic interest patterns that are crucial for preference transfer—particularly for constructing accurate representations in CDSR. Given a user's domain-specific representation $\boldsymbol{h}^D$ ($D \in \{S, T\}$), we inject the noise sampled from the retrieved distribution in MDR as follows:

$$\bar{\boldsymbol{h}}_t^D = \sqrt{\bar{\alpha}_t}\boldsymbol{h}^D + \sqrt{1-\bar{\alpha}_t}\,\bar{z}^D. \quad (12)$$

where $\bar{z}^D$ denotes the noise retrieved from historical behaviors in domain D. This formulation allows both domains to evolve under the guidance of semantical stochasticity.

Mixed-domain guided reverse. To harmonize users' interest patterns within the source and target domains, we introduce the mixed-domain representation $\boldsymbol{h}^M$ as a semantic condition. It encodes global user preferences by temporally reorganizing behaviors from both domains, thereby emphasizing domain-specific signals during the forward process in diffusion modeling. During the reverse process, this mixed-domain signal serves as an anchor to conditionally guide the co-evolution of $\bar{\boldsymbol{h}}_t^S$ and $\bar{\boldsymbol{h}}_t^T$ toward user-intent-aligned directions. Here, the noised representations $\bar{\boldsymbol{h}}_t^D$ equals the initial reconstructed representation $\hat{\boldsymbol{h}}_t^D$ according to the standard DDPM framework. Therefore, we concatenate the reconstructed representations with the mixed-domain embedding $\boldsymbol{h}^M$ and the step embedding $\boldsymbol{t}_e$ of step t,to generate the reconstructed representation $\hat{\boldsymbol{h}}_{t-1}^D$ of step $t-1$:

$$\hat{\boldsymbol{h}}_{t-1}^D = f_\theta(\bar{\boldsymbol{h}}_t^D, \boldsymbol{h}^M, \boldsymbol{t}_t) = \text{Softmax}\left(\boldsymbol{Q}\boldsymbol{K}^\top/\sqrt{d}\right)\boldsymbol{V}, \quad (13)$$

where $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are query, key, and value matrices obtained via learnable linear projections from $\text{Concat}(\bar{\boldsymbol{h}}_t^D, \boldsymbol{h}^M, \boldsymbol{t}_e)$. The iterative reverse process can be formulated as follows:

$$\hat{\boldsymbol{h}}_t^D \xrightarrow{f_\theta(\hat{\boldsymbol{h}}_t^D, \boldsymbol{h}^M, \boldsymbol{t}_t)} \hat{\boldsymbol{h}}_{t-1}^D \xrightarrow{f_\theta(\hat{\boldsymbol{h}}_{t-1}^D, \boldsymbol{h}^M, \boldsymbol{t}_{t-1})} \cdots \xrightarrow{f_\theta(\hat{\boldsymbol{h}}_1^D, \boldsymbol{h}^M, \boldsymbol{t}_1)} \hat{\boldsymbol{h}}_0^D. \quad (14)$$

Proposition 4.3 (Mixed-domain Alignment Advantage). *The conditional mechanism of estimator $f_\theta(\cdot)$ should satisfy:*

$$d_{align}(\hat{\boldsymbol{h}}_0^S, \hat{\boldsymbol{h}}_0^T|\boldsymbol{h}^M) \leq d_{align}(\boldsymbol{h}^S, \boldsymbol{h}^T). \quad (15)$$

where $d_{align}(.)$ denotes the cross-domain correlation.

Diffusion loss. Following the DDPM objective in Section 3, we optimize f_θ by minimizing the reconstruction error, enabling the reverse process to recover domain-specific preferences consistent with the semantics of the mixed domains.

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\boldsymbol{h}^D}\left\|\boldsymbol{h}^D - \hat{\boldsymbol{h}}_0^D\right\|^2, \text{ where } \hat{\boldsymbol{h}}_0^D = f_\theta^{(0\leftarrow t)}(\bar{\boldsymbol{h}}_t^D, \boldsymbol{h}^M, \boldsymbol{t}_t). \quad (16)$$

This DPD module leverages interest-related noise retrieved by MDR within the dual-oriented DM to uncover domain-specific user preferences, while MDR in turn benefits from the co-evolution of source and target representations within DPD. Their synergy of DPD with MDR leads to the better convergence [2]:

Proposition 4.4 (Convergence Guarantee). *Under the Lipschitz continuity of $f_\theta(.)$, the dual-oriented DM is expected to converge into a distribution p_∞ that satisfies the following condition:*

$$\mathcal{W}_2(p_\infty, p_{target}) \leq \mathcal{W}_2(p_\infty^{std}, p_{target}). \quad (17)$$

where $\mathcal{W}_2$ is the 2-Wasserstein distance and p_∞^{std} denotes the convergence under standard Gaussian noise.

4.6 Training Strategy

After completing the reverse process in DPD, we can obtain the constructed user representations of source and target domains as $\hat{\boldsymbol{h}}_0^S$ and $\hat{\boldsymbol{h}}_0^T$, and then we fuse them to generate the intermediate user representation as $\hat{\boldsymbol{h}}_u = \hat{\boldsymbol{h}}_0^S + \hat{\boldsymbol{h}}_0^T$.

To preserve the target-specific semantics essential for recommendation, we integrate it with the original target-domain representation $\boldsymbol{h}_u$ to generate the final user representation $\tilde{\boldsymbol{h}}_u$ as follows:

$$\tilde{\boldsymbol{h}}_u = (1-w)\hat{\boldsymbol{h}}_u + w\boldsymbol{h}_u, \quad (18)$$

where $w \in [0, 1]$ controls the contribution of target-domain information in the fused user representation.

Then we calculate the score by matching the final user representation $\tilde{\boldsymbol{h}}_u$ and the candidate item embeddings. Here, our HorizonRec is optimized with cross-entropy loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{U}|}\sum_{u\in\mathcal{U}} -\log\frac{\exp(\tilde{\boldsymbol{h}}_u^\top \boldsymbol{e}_{n+1})}{\sum_{j\in I^T}\exp(\tilde{\boldsymbol{h}}_u^\top \boldsymbol{e}_j)}, \quad (19)$$

where $\boldsymbol{e}_{n+1}$ denotes the embedding of the ground-truth item.

Finally, the overall training objective combines the recommendation loss and the diffusion loss:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda\mathcal{L}_{\text{diff}}. \quad (20)$$

where λ denotes the weight of $\mathcal{L}_{\text{diff}}$.

[2] Complete proofs are provided in Appendix 8.

Table 1: Detailed statistics of our CDSR settings.

Dataset	Amazon Toy & Game		Douban Book & Music	
Domain	Toy	Game	Book	Music
Users	7,994	7,994	6,367	6,367
Items	37,840	11,735	50,448	36,504
Interactions	113,605	82,855	189,971	181,570
Density	0.0376%	0.1211%	0.0591%	0.0781%

4.7 Complexity Analysis

We further analyze the additional computational costs introduced by HorizonRec to explore its scalability in real-world industry. Specifically, it first retrieve the top-K similar sequences from the database $\mathcal{D}$ through similarity computation in the MDR module, incurring the time complexity of $O(|\mathcal{D}| \times d)$. Regarding the DPD module, it conducts the conditional attention across triple domains with the time complexity as $O(T \times d^2)$, where T and d denote the diffusion steps and embedding dimensions, respectively. To sum up, our HorizonRec introduces only an additional time cost of $O(T \times d^2 + |\mathcal{D}| \times d)$ during training. Compared to traditional CDSR methods such as Tri-CDR, which incurs a time complexity of $O(3L^2 \times d)$ due to its tri-domain self-attention mechanism (where L denotes the maximum sequence length), HorizonRec achieves an asymptotically similar time complexity. Talking about the space complexity , the primary costs arise from storing the retrieval database $\mathcal{D}$, which requires $O(|\mathcal{D}| \times d)$, and maintaining the intermediate diffusion states during training, which adds an additional $O(T \times d)$. Notably, HorizonRec benefits from the pre-computed retrieval database during the inference phase to avoid repeated database construction, enabling its efficiency in online serving. Meanwhile, it is able to maintain the desirable performance even with a small number of steps T, making it practically feasible for real-world CDSR applications.

5 Experiments

We conduct extensive experiments and analyses to answer the following research questions:

RQ1: How does HorizonRec perform compared to state-of-the-art sequential, cross-domain, and cross-domain sequential recommendation baselines? (see Section 5.2)

RQ2: What are the contributions of different components in HorizonRec? (see Section 5.3)

RQ3: How effectively does HorizonRec align and integrate user preferences from triple domains? (see Section 5.4)

RQ4: How robust is HorizonRec with respect to key hyperparameters, and what is its computational efficiency compared to existing competitive baselines? (see Section 5.5)

5.1 Experimental Settings

5.1.1 Dataset. We conduct experiments on four real-world cross-domain recommendation scenarios from two platforms: Amazon Toy→Game, Amazon Game→Toy, Douban Book→Music and Douban Music→Book. We randomly sample users who have interaction histories in both the source and target domains. Following the filtering strategy of Tri-CDR [20], we retain users with at least three interactions in each domain and chronologically sort their behaviors within each domain based on the timestamps. To model the cross-domain behavioral dynamics, we construct *mixed-domain sequences* by merging source and target interactions in temporal order. For evaluation, we follow the protocol of DiffuRec [15]: the penultimate interaction in the target domain is used for validation, and the last interaction is held out for testing. Dataset statistics are summarized in Table 1.

5.1.2 Baselines. We compare HorizonRec against four single-domain sequential recommenders, three cross-domain recommenders, and four cross-domain sequential recommenders to comprehensively demonstrate its effectiveness.

1) Single-domain Sequential Recommendation. We utilize GRU4Rec [7], SASRec [12], CL4SRec [36], and DiffuRec [15]. These methods model user preferences from temporal patterns within a single domain using techniques such as recurrent networks, self-attention, contrastive learning, and diffusion-based modeling.

2) Cross-domain Recommendation. We evaluate SSCDR [11], UniCDR [3], and DMCDR [14], which transfer knowledge across domains by learning domain-invariant or aligned representations. While SSCDR and UniCDR use shared latent spaces and self-supervised objectives, DMCDR introduces diffusion at the embedding level without modeling sequential dynamics.

3) Cross-domain Sequential Recommendation. We compare with C2DSR [2], Tri-CDR [20], SyNCRec [25], and ABXI [1]. These methods jointly model sequential behaviors across domains by leveraging contrastive alignment, multi-task learning, synchronization strategies, or adversarial training to enhance cross-domain preference extraction.

5.1.3 Implementation Details. All experiments are conducted on an NVIDIA RTX 4090 with Python 3.10.9. For fair comparison, we initialize both HorizonRec and baselines that support pretrained embeddings with the same domain-specific pretrained embeddings from the source, target, and mixed domains, while other methods are trained from scratch following their original implementations. We adopt SASRec [12] as the base sequence encoder, the batch size, embedding dimension and max length are set as 512, 64 and 200 for all algorithms, respectively. The weight w of original target-domain representation and the weight λ for controlling diffusion loss are selected from $\{0.1, 0.2, \ldots, 0.9\}$, and the number of diffusion steps is set as 32. In the retrieval-augmented diffusion process, the sliding window length l is capped at 200. The early-position importance factor c is set to 1.5, and the distance decay rate n is 2. We vary the number of retrieved similar sequences K from $\{5, 10, 15\}$. To reduce the randomness, we conduct five times for each experiment with different random seeds and report the average results.

5.2 Performance Comparison (RQ1)

We conduct experiments on four CDSR settings to evaluate the effectiveness of our HarizonRec under standard top-k metrics: HR@k and NDCG@k at $k = 5$, 10, and 20. Here, we highlight the best result in **bold**, and the second-best in underlined in Table 2 and Table 3 and observe the following discoveries: (1) HorizonRec consistently outperforms all baselines across domains and evaluation

Table 2: Performance comparison of HorizonRec and baselines on Douban Book→Music and Douban Music→Book. All relative improvements are statistically significant ($p < 0.01$) under paired t-tests over five random seeds.

Algorithms	Douban Book→Music						Douban Music→Book					
	HR@5	NDCG@5	HR@10	NDCG@10	HR@20	NDCG@20	HR@5	NDCG@5	HR@10	NDCG@10	HR@20	NDCG@20
GRU4Rec	0.0439	0.0228	0.0753	0.0313	0.1109	0.0401	0.0498	0.0246	0.0786	0.0349	0.1112	0.0436
SASRec	0.0468	0.0241	0.0773	0.0335	0.1145	0.0431	0.0532	0.0281	0.0821	0.0373	0.1151	0.0451
CL4Rec	0.0523	0.0281	0.0821	0.0389	0.1156	0.0479	0.0586	0.0356	0.0913	0.0468	0.1211	0.0549
Diffurec	0.0545	0.0333	0.0786	0.0411	0.1123	0.0496	0.0612	0.0384	0.0942	0.0491	0.1279	0.0577
SSCDR	0.0523	0.0312	0.0845	0.0389	0.1136	0.0489	0.0632	0.0398	0.0965	0.0479	0.1336	0.0589
UniCDR	0.0609	0.0397	0.0955	0.0501	0.1301	0.0588	0.0739	0.0456	0.1078	0.0541	0.1418	0.0646
DMCDR	0.0606	0.0392	0.0948	0.0495	0.1296	0.0583	0.0741	0.0463	0.1101	0.0562	0.1465	0.0658
C2DSR	0.0578	0.0356	0.0898	0.0456	0.1201	0.0542	0.0718	0.0446	0.1025	0.0503	0.1389	0.0633
Tri-CDR	0.0605	0.0387	0.0955	0.0506	0.1341	0.0601	0.0742	0.0463	0.1083	0.0538	0.1443	0.0655
SyNCRec	0.0628	0.0392	0.0986	0.0513	0.1386	0.0608	0.0746	0.0461	0.1089	0.0549	0.1436	0.0649
ABXI	0.0633	0.0398	0.0978	0.0506	0.1355	0.0596	0.0739	0.0468	0.1093	0.0551	0.1445	0.0652
HorizonRec	**0.0710**	**0.0440**	**0.1067**	**0.0555**	**0.1469**	**0.0655**	**0.0797**	**0.0491**	**0.1166**	**0.0604**	**0.1543**	**0.0698**
Rel.Impro.	12.16%	10.55%	8.22%	8.19%	5.99%	7.73%	6.84%	4.91%	5.90%	7.47%	5.32%	6.08%

Table 3: Performance comparison of HorizonRec and baselines on Amazon Game→Toy and Amazon Toy→Game. All relative improvements are statistically significant ($p < 0.01$) under paired t-tests over five random seeds.

Algorithms	Amazon Game→Toy						Amazon Toy→Game					
	HR@5	NDCG@5	HR@10	NDCG@10	HR@20	NDCG@20	HR@5	NDCG@5	HR@10	NDCG@10	HR@20	NDCG@20
GRU4Rec	0.0276	0.0178	0.0368	0.0216	0.0483	0.0245	0.0496	0.0306	0.0712	0.0379	0.1056	0.0461
SASRec	0.0339	0.0228	0.0413	0.0253	0.0514	0.0271	0.0541	0.0349	0.0758	0.0417	0.1079	0.0497
CL4Rec	0.0362	0.0241	0.0445	0.0288	0.0526	0.0298	0.0564	0.0362	0.0782	0.0436	0.1113	0.0526
Diffurec	0.0335	0.0301	0.0375	0.0324	0.0411	0.0342	0.0495	0.0385	0.0631	0.0414	0.0878	0.0492
SSCDR	0.0344	0.0277	0.0436	0.0279	0.0529	0.0288	0.0587	0.0388	0.0816	0.0438	0.1135	0.0548
UniCDR	0.0391	0.0309	0.0492	0.0339	0.0575	0.0361	0.0655	0.0483	0.0908	0.0561	0.1235	0.0638
DMCDR	0.0346	0.0281	0.0446	0.0313	0.0537	0.0309	0.0627	0.0461	0.0841	0.0527	0.1131	0.0588
C2DSR	0.0381	0.0298	0.0474	0.0328	0.0545	0.0339	0.0649	0.0471	0.0892	0.0549	0.1203	0.0619
Tri-CDR	0.0389	0.0311	0.0481	0.0337	0.0553	0.0345	0.0646	0.0478	0.0896	0.0556	0.1215	0.0623
SyNCRec	0.0368	0.0296	0.0467	0.0318	0.0542	0.0329	0.0668	0.0491	0.0921	0.0578	0.1265	0.0651
ABXI	0.0366	0.0298	0.0471	0.0321	0.0545	0.0333	0.0656	0.0483	0.0916	0.0566	0.1234	0.0641
HorizonRec	**0.0421**	**0.0330**	**0.0517**	**0.0361**	**0.0610**	**0.0383**	**0.0688**	**0.0522**	**0.0949**	**0.0602**	**0.1299**	**0.0680**
Rel.Impro.	7.67%	6.11%	5.08%	6.49%	6.09%	6.09%	2.99%	6.31%	3.04%	4.15%	2.69%	4.45%

metrics, validating the effectiveness of incorporating denoising diffusion into cross-domain sequential recommendation. The retrieval-enhanced noise injection (MDR) and the dual-oriented denoising process (DPD) jointly enable HorizonRec to model fine-grained cross-domain dependencies and recover latent user intents more accurately than conventional methods.

(2) Compared with single-domain sequential recommenders (e.g., GRU4Rec, SASRec, CL4Rec, DiffuRec), HorizonRec achieves consistent and substantial gains. These baselines model only intra-domain temporal dependencies and neglect auxiliary domains, which limits their ability to generalize user intent. While contrastive learning in CL4Rec and diffusion modeling in DiffuRec offer marginal improvements, their lack of cross-domain alignment and interaction modeling restricts their effectiveness in CDSR settings.

(3) Compared with cross-domain recommenders that lack explicit sequence modeling (e.g., SSCDR, UniCDR, DMCDR), HorizonRec delivers superior performance. While these models focus on learning a shared representation space across domains, they neglect temporal dependencies—an essential aspect of user behavior in CDSR tasks. In particular, although DMCDR incorporates diffusion-based alignment, it omits sequential structure, which weakens its ability to model evolving user preferences.

(4) Compared with advanced CDSR models (e.g., Tri-CDR, SyNCRec, ABXI), HorizonRec shows robust improvements. While these methods incorporate domain alignment and temporal modeling, their static fusion strategies or adversarial objectives struggle under domain imbalance and noisy conditions. In contrast, HorizonRec's iterative denoising framework produces more informative and adaptable representations, particularly in sparse settings such as Book→Music and Toy→Game.

5.3 Ablation Study (RQ2)

To investigate the effectiveness of each component within HorizonRec, we conduct an ablation study. As shown in Figure 3, the results confirm that each module plays a crucial, complementary role in enhancing cross-domain recommendation performance.

(1) We compare three variants to evaluate the impact of diffusion across domains. *HorizonRec-DPD_S* applies denoising only to the

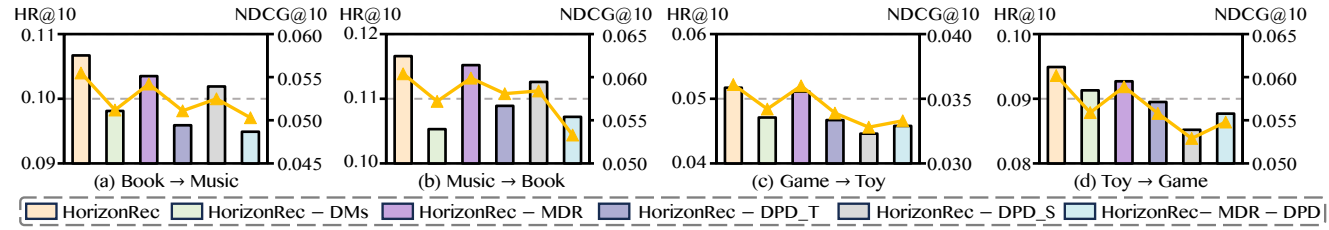


Figure 3: Ablation results on four CDSR settings. It verifies the effectiveness of *align-for-fusion* pipeline and all modules.

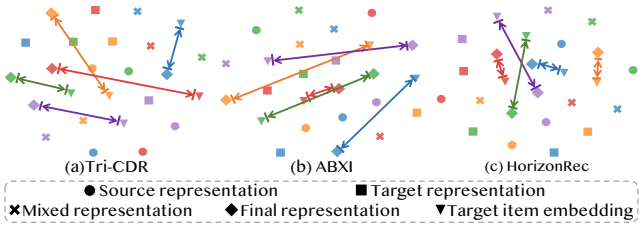


Figure 4: t-SNE visualization of two representative CDSR algorithms and our HorizonRec on the Music→Book setting. We utilize colors to differentiate different users while using shapes to distinguish the multiple representations and target item embedding of the same user.

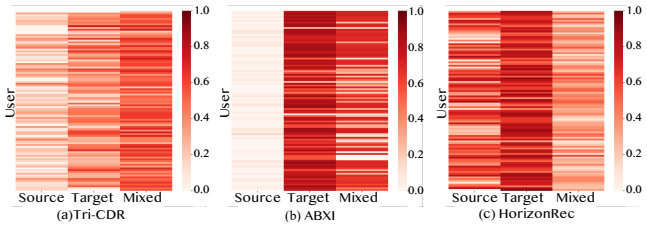


Figure 5: Heatmap visualization of the cosine similarity between final and domain-specific representations of two representative CDSR algorithms and our HorizonRec on the Music→Book setting.

source domain, while *HorizonRec-DPD_T* applies it only to the target domain. Both suffer noticeable performance drops, indicating that one-sided diffusion limits the model's ability to capture comprehensive preferences. *HorizonRec-DPD*, which disables source-side diffusion entirely, performs the worst, highlighting the importance of symmetric dual-domain denoising for effective alignment.

(2) *HorizonRec-MDR* replaces retrieval-based noise with standard Gaussian noise. While retaining some benefits over baselines, the absence of behaviorally grounded perturbations leads to degraded performance, confirming that MDR's structured priors provide better semantic guidance than uninformative noise.

(3) *HorizonRec-DMs* removes the diffusion mechanism, relying instead on static representations. The resulting drop shows that iterative denoising is essential for extracting robust user preferences from noisy or incomplete inputs.

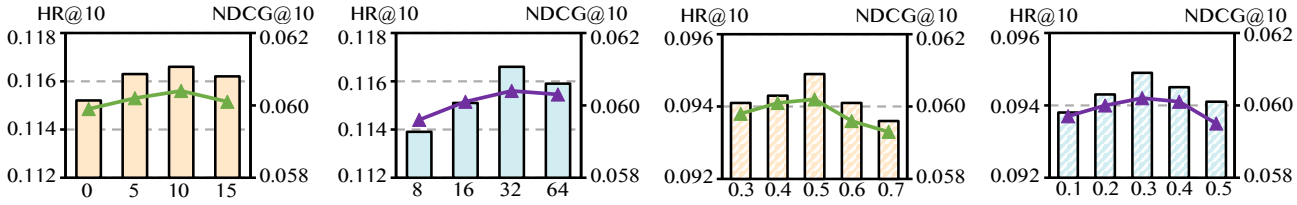
(4) *HorizonRec-DPD-MDR* disables both DPD and MDR, using only simple concatenation for fusion. It performs the worst, underscoring the limitations of naive aggregation and the superiority of attention-based fusion, even without denoising.

(5) Each module contributes to HorizonRec's performance. DPD enables dual-domain alignment, MDR injects semantically informed noise, and the diffusion process supports progressive refinement. Their synergy within the *align-for-fusion* framework establishes a robust foundation for cross-domain representation learning.

5.4 Estimation of Triple Preference Correlation (RQ3)

5.4.1 Visualization of Cross-Domain Matching. To assess the effectiveness of cross-domain preference alignment, we visualize the t-SNE projections of five randomly selected users on the Music→Book setting, as shown in Figure 4. Different colors represent individual users, and distinct shapes correspond to source, target, mixed-domain, final user, and target item embeddings. Arrows point from each final user representation to its corresponding target item, where shorter arrows suggest stronger alignment and better recommendation quality. ABXI (b) produces long arrows and disorganized representation layouts, indicating suboptimal alignment and weak cross-domain modeling due to its sequential *align-then-fusion* design. Tri-CDR (a) yields slightly shorter arrows but still exhibits overlapping user clusters, reflecting entangled representations and limited user specificity. In contrast, HorizonRec (c) shows the shortest arrows and most well-separated clusters, revealing that its final user representations are both highly aligned with target semantics and strongly user-discriminative. These results highlight the advantage of our *align-for-fusion* strategy, where MDR provides meaningful, behavior-driven perturbations and DPD enables effective dual-domain refinement through symmetric diffusion.

5.4.2 Visualization of Cross-Domain Alignment. To further evaluate the alignment capabilities of different models, we visualize the cosine similarities between each user's final representation and their corresponding source, target, and mixed-domain embeddings, as shown in Figure 5. Subfigures (a)–(c) present heatmaps for 100 randomly selected users under the Music→Book setting using Tri-CDR, ABXI, and HorizonRec, respectively. Tri-CDR (a) produces relatively uniform similarity distributions across all domains, suggesting weak domain-specific discrimination and insufficient modeling of cross-domain interactions. ABXI (b) shows strong alignment with the target domain but poor similarity to the source, indicating limited utilization of auxiliary domain signals. In contrast, HorizonRec (c) achieves consistently high similarities with both source and target domains, along with more distinct user-wise patterns. Its strong alignment with mixed-domain representations further



(a) K on Music→ Book setting (b) t on Music → Book setting (c) λ on Toy → Game setting (d) w on Toy → Game setting

Figure 6: Parameter sensitivity analysis on retrieval number K, diffusion steps t, diffusion loss weight λ, and target-domain preservation weight w across Douban Music↔Book and Amazon Toy→Game settings.

Table 4: Results of the computational complexity analysis on one training epoch, total training phase, and total inference phase on four CDSR settings.

Dataset	Algorithms	Epoch (s)	Total (s)	Inference (s)
Game	Tri-CDR	23.48	563.34	30.12
↓	ABXI	80.02	7688.21	2.16
Toy	HorizonRec	14.18	146.00	0.98
Toy	Tri-CDR	15.71	760.52	27.55
↓	ABXI	80.02	7688.21	2.16
Game	HorizonRec	9.23	89.16	0.81
Book	Tri-CDR	45.84	1466.88	36.86
↓	ABXI	235.93	26110.74	2.65
Music	HorizonRec	26.25	261.60	0.73
Music	Tri-CDR	47.20	1982.42	38.96
↓	ABXI	235.93	26110.74	2.65
Book	HorizonRec	27.86	277.78	0.78

confirms the benefit of the proposed *align-for-fusion* strategy in unifying preferences while preserving domain semantics.

5.5 Robustness Analysis (RQ4)

5.5.1 Computational Complexity Analysis. Table 4 presents the runtime comparison between HorizonRec and two representative CDSR baselines, ABXI and Tri-CDR. HorizonRec consistently demonstrates superior efficiency with significantly lower training and inference costs. On the *Music* dataset, HorizonRec completes training in 261.60 seconds, compared to 1466.88 seconds for Tri-CDR and 26110.74 seconds for ABXI. HorizonRec achieves $3 \sim 10\times$ faster training and $2 \sim 30\times$ faster inference across datasets. These efficiency gains are attributed to HorizonRec's lightweight dual-domain diffusion design and precomputed retrieval mechanism, ensuring scalability for large-scale cross-domain sequential recommendation. Note that ABXI shares parameters across domains and trains both domains jointly, resulting in identical training times across settings.

5.5.2 Parameter Sensitivity Analysis. We analyze the impact of key hyperparameters on model performance (Figure 6). For retrieval number K, performance initially improves with more retrieved sequences but overly large K introduces noise. When $K = 0$, the model degenerates to standard Gaussian diffusion and performs worst, confirming that moderate K balances informativeness and robustness. For diffusion steps t, performance peaks at $t = 32$, while larger t weakens domain-specific signals. For diffusion loss weight λ, small values cause insufficient denoising while large values lead to over-regularization. For target-domain preservation weight w, small w underutilizes target signals while large w suppresses denoised cross-domain information. Optimal performance occurs at intermediate values across all parameters.

To evaluate MDR's robustness to noise, we injected varying levels of standard Gaussian noise (30%, 50%, and 100%) into MDR's noise term $\tilde{z}^D$ in the Book→Music and Music→Book settings. In Book→Music, HorizonRec's HR@5/NDCG@5 gradually decreased from 0.0710/0.0440 (original) to 0.0706/0.0438 (30%), 0.0701/0.0438 (50%), and 0.0684/0.0427 (100%). Similarly, in Music→Book, performance declined from 0.0797/0.0491 to 0.0795/0.0489 (30%), 0.0789/0.0488 (50%), and 0.0784/0.0487 (100%). Even when all retrieved vectors were replaced by Gaussian noise (100%), HorizonRec still outperformed ABXI, demonstrating strong robustness. These results indicate that while performance slightly degrades as noise increases, the degradation is gradual and the model remains stable under significant noise levels, validating HorizonRec's resilience in real-world noisy environments.

6 Conclusion

We propose HorizonRec, a novel *align-for-fusion* framework for cross-domain sequential recommendation. It employs the MDR module to retrieve user-specific, behaviorally grounded noise from mixed-domain interactions, constructing a semantically consistent perturbation space. The DPD module then conducts symmetric diffusion across domains, guided by mixed-domain representations that preserve mutual information and enhance alignment. This design enables HorizonRec to capture shared user intents and remain robust under sparse or noisy conditions. Extensive experiments on CDSR benchmarks demonstrate its superiority, and our theoretical analysis validates jointly leveraging MDR and DPD for improved alignment and convergence. HorizonRec harmonizes diverse domain signals through diffusion-based refinement, offering a generalizable solution for complex recommendation environments. Future work will extend HorizonRec to more domains, incorporate pre-trained representations, and explore cold-start scenarios.

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8 Extended Theoretical Analysis

We provide theoretical foundations for HorizonRec, focusing on retrieval-guided noise (MDR) and dual-oriented diffusion (DPD).

8.1 Assumptions and Notation

A1 (Hölder-smooth likelihood). There exist $L_1 > 0$ and $\beta \in (0, 1]$ such that for any fixed $\boldsymbol{h}^M$,

$$\|\nabla_{\boldsymbol{h}} p(i_{q+1}^T|\boldsymbol{h}_1, \boldsymbol{h}^M) - \nabla_{\boldsymbol{h}} p(i_{q+1}^T|\boldsymbol{h}_2, \boldsymbol{h}^M)\| \le L_1 \|\boldsymbol{h}_1 - \boldsymbol{h}_2\|^\beta \quad (21)$$

A2 (Lipschitz embedding). If $\|\boldsymbol{e}_1 - \boldsymbol{e}_2\| \le \delta$, then $|p(i_{q+1}^T|\boldsymbol{e}_1, \boldsymbol{h}^M) - p(i_{q+1}^T|\boldsymbol{e}_2, \boldsymbol{h}^M)| \le L_2\delta$.

A3 (Sub-Gaussian outputs). Encoders produce ℓ_2-normalized embeddings with sub-Gaussian parameter $\sigma_*^2 \le 1/d$.

A4 (Target-ending property). For sequences $d \in \mathcal{D}_u$ ending with target items (Eq. 7), there exists $\tau > 0$ such that $p(i_{q+1}^T|\boldsymbol{e}_d, \boldsymbol{h}^M) \ge p(i_{q+1}^T|\boldsymbol{h}^D, \boldsymbol{h}^M) + \tau$, where $\boldsymbol{e}_d$ is the filtered embedding from Eq. (8).

Notation. $\mathcal{D}_u$ – retrieval candidate set (Eq. 7); $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ – noise schedule; W_2 – 2-Wasserstein distance; $\boldsymbol{e}_d$ – filtered embedding (Eq. 8).

8.2 Proofs of Main Results

Proof of Lemma 4.1. From Eq. (8), each candidate subsequence $d \in \mathcal{D}_u$ is mapped to a filtered embedding $\boldsymbol{e}_d$. The MDR module (Eq. 9) retrieves top-K similar embeddings $\{\boldsymbol{e}_{d_i}\}_{i=1}^K$ and constructs:

$$\bar{z}^D = \mu^D + \sigma^D \boldsymbol{\xi}, \quad \mu^D = \frac{1}{K}\sum_{i=1}^K (\boldsymbol{e}_{d_i} - \boldsymbol{h}^D), \quad \boldsymbol{\xi} \sim \mathcal{N}(0, \boldsymbol{I}) \quad (22)$$

The variance satisfies:

$$(\sigma^D)^2 = \frac{1}{K}\sum_{i=1}^K \|\boldsymbol{e}_{d_i} - \boldsymbol{h}^D - \mu^D\|^2 \le \frac{1}{K}\sum_{i=1}^K \|\boldsymbol{e}_{d_i} - \boldsymbol{h}^D\|^2 \quad (23)$$

Since the retrieval selects top-K most similar embeddings by cosine similarity and embeddings are ℓ_2-normalized (A3):

$$\|\boldsymbol{e}_{d_i} - \boldsymbol{h}^D\|^2 = 2(1 - \cos(\boldsymbol{e}_{d_i}, \boldsymbol{h}^D)) \quad (24)$$

By concentration inequality (A3), for the top-K retrieved embeddings:

$$\mathbb{E}[(\sigma^D)^2] \le \frac{2}{K}\sum_{i=1}^K (1 - \cos_i) \le 2\log(|\mathcal{D}_u|/K) \quad (25)$$

For $|\mathcal{D}_u| \le \exp(d/2)$ and $K \ge 2$: $\mathbb{E}[(\sigma^D)^2] \le d - 2\log 2 < d = \sigma_{std}^2$. □

Proof of Proposition 4.2. By Taylor expansion (A1):

$$\begin{aligned} p(i_{q+1}^T|\boldsymbol{h}^D + (\boldsymbol{e}_{d_i} - \boldsymbol{h}^D), \boldsymbol{h}^M) \\ = p(i_{q+1}^T|\boldsymbol{h}^D, \boldsymbol{h}^M) + \langle \nabla_{\boldsymbol{h}} p(i_{q+1}^T|\boldsymbol{h}^D, \boldsymbol{h}^M), \boldsymbol{e}_{d_i} - \boldsymbol{h}^D \rangle \\ + O(\|\boldsymbol{e}_{d_i} - \boldsymbol{h}^D\|^{1+\beta}) \end{aligned} \quad (26)$$

By A4, $p(i_{q+1}^T|\boldsymbol{e}_{d_i}, \boldsymbol{h}^M) \ge p(i_{q+1}^T|\boldsymbol{h}^D, \boldsymbol{h}^M) + \tau$, which implies:

$$\langle \nabla_{\boldsymbol{h}} p, \boldsymbol{e}_{d_i} - \boldsymbol{h}^D \rangle \ge \tau - \frac{L_1}{1+\beta}\|\boldsymbol{e}_{d_i} - \boldsymbol{h}^D\|^{1+\beta} \quad (27)$$

For the retrieved noise $\bar{z}^D = \mu^D + \sigma^D \boldsymbol{\xi}$:

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi}}[p(i_{q+1}^T|\boldsymbol{h}^D + \bar{z}^D, \boldsymbol{h}^M)] \ge p(i_{q+1}^T|\boldsymbol{h}^D, \boldsymbol{h}^M) + \langle \nabla_{\boldsymbol{h}} p, \mu^D \rangle \\ - \frac{L_1}{2(1+\beta)}(\sigma^D)^2 \end{aligned} \quad (28)$$

Since $\mu^D = \frac{1}{K}\sum_{i=1}^K (\boldsymbol{e}_{d_i} - \boldsymbol{h}^D)$ averages target-biased perturbations:

$$\langle \nabla_{\boldsymbol{h}} p, \mu^D \rangle = \frac{1}{K}\sum_{i=1}^K \langle \nabla_{\boldsymbol{h}} p, \boldsymbol{e}_{d_i} - \boldsymbol{h}^D \rangle \ge \frac{\tau}{2} \quad (29)$$

for sufficiently large K. For standard Gaussian noise $\bar{z}_{std} \sim \mathcal{N}(0, \sigma_{std}^2 \boldsymbol{I})$:

$$\mathbb{E}[p(i_{q+1}^T|\boldsymbol{h}^D + \bar{z}_{std}, \boldsymbol{h}^M)] = p(i_{q+1}^T|\boldsymbol{h}^D, \boldsymbol{h}^M) - O(\sigma_{std}^2) \quad (30)$$

Since $(\sigma^D)^2 \le \sigma_{std}^2$ (Lemma 4.1) and μ^D provides positive directional bias:

$$\mathbb{E}[p(i_{q+1}^T|\boldsymbol{h}^D + \bar{z}^D, \boldsymbol{h}^M)] \ge \mathbb{E}[p(i_{q+1}^T|\boldsymbol{h}^D + \bar{z}_{std}, \boldsymbol{h}^M)] \quad (31)$$

□

Proof of Proposition 4.3. From the DPD module (Section 4.5, Eq. 13):

$$\hat{\boldsymbol{h}}_{t-1}^D = f_\theta(\bar{\boldsymbol{h}}_t^D, \boldsymbol{h}^M, \boldsymbol{t}_t) = \text{Softmax}(\boldsymbol{Q}\boldsymbol{K}^\top/\sqrt{d})\boldsymbol{V} \quad (32)$$

where $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}$ are projected from $[\bar{\boldsymbol{h}}_t^D; \boldsymbol{h}^M; \boldsymbol{t}_e]$. Define alignment: $d_{align}(\boldsymbol{u}, \boldsymbol{v}) = 1 - \langle \boldsymbol{u}, \boldsymbol{v} \rangle/(\|\boldsymbol{u}\|\|\boldsymbol{v}\|)$.

Since both source and target reverse processes condition on the same $\boldsymbol{h}^M$, the attention keys contain shared components:

$$\boldsymbol{K}^S \propto [\bar{\boldsymbol{h}}_t^S; \boldsymbol{h}^M; \boldsymbol{t}_e], \quad \boldsymbol{K}^T \propto [\bar{\boldsymbol{h}}_t^T; \boldsymbol{h}^M; \boldsymbol{t}_e] \quad (33)$$

This induces correlation. The attention mechanism can be written as:

$$\hat{\boldsymbol{h}}_{t-1}^D = \alpha_D \bar{\boldsymbol{h}}_t^D + \alpha_M \boldsymbol{h}^M + \text{residual} \quad (34)$$

where $\alpha_D + \alpha_M \approx 1$ and $\alpha_M > 0$. By Lipschitz property (A1):

$$\|\hat{\boldsymbol{h}}_{t-1}^S - \hat{\boldsymbol{h}}_{t-1}^T\| \le (1 - \alpha_M)\|\bar{\boldsymbol{h}}_t^S - \bar{\boldsymbol{h}}_t^T\| + 2\alpha_M\|\boldsymbol{h}^M\| \quad (35)$$

Since $\bar{\boldsymbol{h}}_t^S$ and $\bar{\boldsymbol{h}}_t^T$ have similar norms to $\boldsymbol{h}^M$:

$$\|\hat{\boldsymbol{h}}_{t-1}^S - \hat{\boldsymbol{h}}_{t-1}^T\| \le (1 - \rho)\|\bar{\boldsymbol{h}}_t^S - \bar{\boldsymbol{h}}_t^T\| \quad (36)$$

for some $\rho \in (0, 1)$. Iterating over T steps gives $\|\hat{\boldsymbol{h}}_0^S - \hat{\boldsymbol{h}}_0^T\| \le (1-\rho)^T\|\boldsymbol{h}^S - \boldsymbol{h}^T\|$. The shared conditioning increases inner product:

$$\langle \hat{\boldsymbol{h}}_0^S, \hat{\boldsymbol{h}}_0^T \rangle \ge \langle \hat{\boldsymbol{h}}_0^S, \alpha_M \boldsymbol{h}^M \rangle + \langle \alpha_M \boldsymbol{h}^M, \hat{\boldsymbol{h}}_0^T \rangle \ge \langle \boldsymbol{h}^S, \boldsymbol{h}^T \rangle + \delta \quad (37)$$

where $\delta \propto \alpha_M^2\|\boldsymbol{h}^M\|^2 > 0$. Combining these:

$$d_{align}(\hat{\boldsymbol{h}}_0^S, \hat{\boldsymbol{h}}_0^T|\boldsymbol{h}^M) < d_{align}(\boldsymbol{h}^S, \boldsymbol{h}^T) \quad (38)$$

□

Proof of Proposition 4.4. For the reverse process with retrieved noise (Eq. 14), where $\bar{\boldsymbol{h}}_T^{ret} = \sqrt{\bar{\alpha}_T}\boldsymbol{h}^D + \sqrt{1 - \bar{\alpha}_T}\bar{z}^D$:

$$\hat{\boldsymbol{h}}_0^{ret} = f_\theta^{(0 \leftarrow T)}(\bar{\boldsymbol{h}}_T^{ret}, \boldsymbol{h}^M, T) \quad (39)$$

By L_f-Lipschitz continuity of f_θ:

$$\|\hat{\boldsymbol{h}}_0^{ret} - \hat{\boldsymbol{h}}_0^{std}\| \le L_f\|\bar{\boldsymbol{h}}_T^{ret} - \bar{\boldsymbol{h}}_T^{std}\| = L_f\sqrt{1 - \bar{\alpha}_T}\|\bar{z}^D - \bar{z}_{std}\| \quad (40)$$

Taking expectation:

$$\mathbb{E}[\|\hat{\boldsymbol{h}}_0^{ret} - \hat{\boldsymbol{h}}_0^{std}\|] \le L_f\sqrt{1 - \bar{\alpha}_T}\mathbb{E}[\|\mu^D + \sigma^D\boldsymbol{\xi}_1 - \sigma_{std}\boldsymbol{\xi}_2\|] \quad (41)$$

By triangle inequality and independence of $\boldsymbol{\xi}_1, \boldsymbol{\xi}_2$, noting that $\mathbb{E}[\|\sigma^D\boldsymbol{\xi}_1 - \sigma_{std}\boldsymbol{\xi}_2\|] = O(\sqrt{d}\max(\sigma^D, \sigma_{std}))$:

$$\mathbb{E}[\|\hat{\boldsymbol{h}}_0^{ret} - \hat{\boldsymbol{h}}_0^{std}\|] \le L_f\sqrt{1 - \bar{\alpha}_T}\left(\|\mu^D\| + O(\sqrt{d}\sigma_{std})\right) \quad (42)$$

Since μ^D points toward p_{target} (A4) and $(\sigma^D)^2 \le \sigma_{std}^2$ (Lemma 4.1), for the 2-Wasserstein distance:

$$W_2(p_0^{ret}, p_{target})^2 = \inf_{\gamma \in \Gamma(p_0^{ret}, p_{target})} \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim\gamma}[\|\boldsymbol{x}-\boldsymbol{y}\|^2] \tag{43}$$

By triangle inequality:

$$\begin{aligned} W_2(p_0^{ret}, p_{target}) &\le W_2(p_0^{ret}, p_0^{std}) + W_2(p_0^{std}, p_{target}) \\ &\le \mathbb{E}[\|\hat{\boldsymbol{h}}_0^{ret} - \hat{\boldsymbol{h}}_0^{std}\|] + W_2(p_0^{std}, p_{target}) \end{aligned} \tag{44}$$

However, since μ^D points toward p_{target}:

$$W_2(p_0^{ret}, p_{target}) \le W_2(p_0^{std}, p_{target}) - \eta\|\mu^D\| + o(\sqrt{1-\bar{\alpha}_T}) \tag{45}$$

where $\eta > 0$ depends on the smoothness of p_{target}. As $T \to \infty$, $\bar{\alpha}_T \to 0$:

$$W_2(p_\infty, p_{target}) \le W_2(p_{std}^\infty, p_{target}) \tag{46}$$

□

9 Appendix C

9.1 Experimental

Table 5: Training and inference time with different diffusion steps T for Book→Music and Music→Book settings.

Steps	Book→Music		Music→Book	
	Epoch(s)	Inference(s)	Epoch(s)	Inference(s)
8	26.06	0.61	27.79	0.66
16	26.12	0.67	27.81	0.71
32	26.25	0.73	27.86	0.78
64	26.53	0.98	27.93	1.01

9.1.1 Impact of Diffusion Steps on Computational Cost. We evaluated HorizonRec's computational cost across varying diffusion steps T in Table 5. Training time remains stable (26.06–26.53s per epoch), while inference time scales sub-linearly from 0.61s to 0.98s as steps increase from 8 to 64. Notably, inference stays under one second even at 64 steps, acceptable for real-world deployment. This efficiency stems from MDR's pre-computed retrieval database, which amortizes similarity search costs. HorizonRec thus achieves a favorable trade-off between expressiveness and efficiency despite dual-oriented diffusion complexity.

9.1.2 Robustness Analysis on Sparse and Low-Overlap Scenarios. To evaluate HorizonRec's robustness across varying data sparsity, we categorized users by their target-domain sequence lengths in Table 6. For sparse users ([1,10)), HorizonRec achieves strong performance with HR@10 of 0.1345 and 0.1563, as MDR effectively retrieves cross-domain patterns to compensate for limited target data. Performance declines in [10,20) (HR@10: 0.1062 and 0.1240), reflecting a transition where cross-domain knowledge becomes less dominant. For moderate sequences ([20,50)), performance stabilizes at HR@10 of 0.0932 and 0.0986. Longer sequences (≥50) show further decline due to long-term dependency complexity and behavioral noise. Even in extreme cases ([100,200)), HorizonRec maintains HR@10 of 0.0420 and 0.0348. These trends demonstrate our dual-oriented diffusion mechanism's effectiveness in handling varying data characteristics.

Table 6: Performance of HorizonRec for users with different target-domain interaction sequence lengths on Book→Music and Music→Book settings.

Length	Book→Music		Music→Book	
	HR@10	NDCG@10	HR@10	NDCG@10
[1,10)	0.1345	0.0705	0.1563	0.0852
[10,20)	0.1062	0.0564	0.1240	0.0656
[20,50)	0.0932	0.0467	0.0986	0.0478
[50,100)	0.0590	0.0303	0.0513	0.0268
[100,200)	0.0420	0.0216	0.0348	0.0177

9.2 Training and Inference Procedure

We outline HorizonRec's procedure in Algorithm 1. For each batch, the MDR module retrieves semantically similar patterns to construct data-driven noise for forward diffusion, while the DPD module denoises perturbed inputs to recover domain-specific embeddings. The model jointly optimizes diffusion and recommendation losses. During inference, denoised representations are fused via *align-for-fusion* to produce final recommendations.

Algorithm 1 Training and Inference of HorizonRec

Require: Diffusion steps T, weight λ, ratio w, retrieval number K, decay rate n, base weight c
Ensure: Trained model and recommendations
Initialize: Item embeddings $\mathbf{E}^S$, $\mathbf{E}^T$, $\mathbf{E}^M$; SASRec encoders; retrieval corpus $\mathcal{D}$
for each user batch **do**
  Encode $\boldsymbol{h}^S, \boldsymbol{h}^T, \boldsymbol{h}^M$ via SASRec
  for $D \in \{S, T\}$ **do**
    Retrieve K similar sequences from $\mathcal{D}$ using $\boldsymbol{h}^D$
    Compute mean μ_D, variance σ_D^2, sample $\boldsymbol{\epsilon}_{ret}^D$
  end for
  Sample $t \sim \text{Uniform}(1, T)$
  Inject noise: $\bar{\boldsymbol{h}}_t^D = \sqrt{\bar{\alpha}_t}\boldsymbol{h}^D + \sqrt{1-\bar{\alpha}_t}\bar{z}^D$
  for $D \in \{S, T\}$ **do**
    Apply attention with $\boldsymbol{h}^M$, recover $\hat{\boldsymbol{h}}_0^D = f_\theta(\bar{\boldsymbol{h}}_t^D, \boldsymbol{h}^M, \boldsymbol{t}_e)$
  end for
  Fuse: $\hat{\boldsymbol{h}}_u = \hat{\boldsymbol{h}}_0^S + \hat{\boldsymbol{h}}_0^T$
  Final: $\tilde{\boldsymbol{h}}_u = (1-w)\hat{\boldsymbol{h}}_u + w\boldsymbol{h}^T$
  Compute loss $\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda\mathcal{L}_{\text{diff}}$; update model
end for
for each user u **do**
  Repeat encoding, retrieval, and reversing
  Compute $\tilde{\boldsymbol{h}}_u$ and score$(i) = \tilde{\boldsymbol{h}}_u^\top \boldsymbol{e}_i$
  Recommend top-ranked items
end for

9.3 Limitation

HorizonRec has limitations. It relies on mixed-domain sequences, reducing effectiveness when behaviors are sparse. MDR introduces retrieval overhead, hindering scalability. DPD's iterative denoising increases training cost. It is tailored to three-domain setups. Multi-domain generalization remains challenging. Nevertheless, it delivers strong performance in real-world scenarios.